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Multi-timescale optimization scheduling of interconnected data centers based on model predictive control

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Abstract With the promotion of “dual carbon” strategy, data center (DC) access to high-penetration renewable energy sources (RESs) has become a trend in the industry. However, the uncertainty of RES poses challenges to the safe and stable operation of DCs and power grids. In this paper, a multi-timescale optimal scheduling model is established for interconnected data centers (IDCs) based on model predictive control (MPC), including day-ahead optimization, intraday rolling optimization, and intraday real-time correction. The day-ahead optimization stage aims at the lowest operating cost, the rolling optimization stage aims at the lowest intraday economic cost, and the real-time correction aims at the lowest power fluctuation, eliminating the impact of prediction errors through coordinated multi-timescale optimization. The simulation results show that the economic loss is reduced by 19.6%, and the power fluctuation is decreased by 15.23%.

Keywords model predictive control, interconnected data center, multi-timescale, optimized scheduling, distributed power supply, landscape uncertainty

1 Introduction

In recent years, data center (DC), as an important part of the Internet and information technology, have been developing rapidly and expanding in scale [1–3]. However, the ensuing problem of high energy consumption in DCs cannot be ignored [4]. In the context of the “dual carbon” strategy, a high percentage of renewable energy sources (RESs) access has become an important means of reducing operational energy consumption, improving energy utilization efficiency,

and promoting the green and low-carbon transformation of DC. In addition, the uncertainty of RES poses a major challenge to optimal scheduling and energy management in DC [5–7]. Therefore, it is of great significance to fully consider the uncertainty problem of renewable energy and make use of the flexible resources of DCs for optimal scheduling in order to reduce the operating cost of DCs and improve the stability of DC operation.

Distinguished from flexible loads such as air conditioning and electric heating, DC loads, as typical high-energy loads, not only flexibly regulate the load characteristics from time to time, but also achieve the spatial transfer of electric loads through the transfer of arithmetic power, which is a high-quality load regulation resource [8–10]. In addition to servers, DCs have other auxiliary equipment, such as cooling, energy storage (ES), generators, and RES power supply, which bring flexibility to the operation [11–14]. To give full play to the advantages of the DC, it is necessary to formulate a reasonable and effective scheduling strategy to coordinate the operation of multi-equipment.

Hitherto, the research on DC operation optimization has mainly focused on flexibility in mobilizing resources, energy forms, and ES devices [15]. Yuan et al. [16,17] took into account the spatial diversity of Internet service provider (ISP) bandwidth prices, electricity prices, and renewable green energy availability, and made full use of the spatio-temporal flexibility of DC workloads for optimal scheduling, which effectively reduced the energy cost and improved the capacity of RES consumption.

In Cioara et al. [18], a flexible mechanism and model was defined for each component of a DC (electrical cooling system, IT workloads, ES, and diesel generators). Optimization measures such as load time shifting, alternative use of non-electrical cooling equipment (e.g., thermal storage), or charging and discharging of ES were used to increase the capacity of RES consumption. In Chen et al. [19], a system framework was proposed to integrate RES, distributed storage units, cooling facilities, and dynamic pricing into the workloads and energy

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management tasks of a DC network, which could efficiently utilize multiple resources to improve the energy efficiency and sustainability of a DC. In Wang et al. [20], a mixed-integer linear programming model was developed to solve the day-ahead resource planning problem with the objective of minimizing the energy cost and carbon emission cost by considering conventional generators, renewable resources, and ES systems.

The above optimization controls are all based on load and RES forecasting information accurately. However, in actual operation, the predicted values of RES and load deviate from the real values, which brought challenges to system operation control [21]. To deal with the uncertainty in prediction, methods such as robust planning [22] and stochastic planning [23–25] had been applied.

Most of the above methods adopt advance scheduling strategies for multiple time periods. Although a rolling optimization method can be adopted for rolling scheduling, the above scheduling models are all approximate models, which cannot fully meet the demand for online optimization and adjustment in the actual operation of the system. The model predictive control (MPC) method based on the idea of rolling optimization and feedback correction can better solve the problem of inaccurate prediction, with a better robustness. In Paul et al. [26], an MPC-based energy-aware scheduling algorithm was proposed to address the resource allocation problem of deferrable jobs in the tiered architecture of DCs, which effectively reduced electricity bills and improved RES consumption. Wu et al. [27] used predictive data and prediction error distribution for real-time energy management of DCs based on the MPC algorithms, taking into account factors such as RES, dynamic tariffs, batteries, thermal storage tanks, and delayed execution of batch jobs. In Zhu et al. [28], a stochastic MPC based multi-temporal optimal scheduling method for DC microgrids were proposed for two phases, i.e., day-ahead and intraday. Intraday scheduling made use of the rolling optimization and feedback correction functions of MPC to correct deviations in load and RES output, and adjusted the day-ahead scheduling plan in real time, which ensured the effectiveness of the day-ahead plan and the stability of the system operation. Although the above studies have enhanced the adaptability to the inaccuracy of prediction information by using the MPC method, they do not take into account the error between the prediction information and the small-timescale operation information, and thus cannot meet the demand of the small-timescale dynamic adjustment during the operation of the system.

Accordingly, in this paper multiple flexibility resources are considered for optimal scheduling of interconnected DCs on multiple timescales. This paper is contributive because it introduces the interconnected data center (IDC) system structure and energy consumption

characteristics. In addition, considering the uncertainty of intraday scenery output and the short-sighted effect of real-time optimization, it proposes a multi-timescale optimal scheduling model for IDC, i.e., a day-ahead optimal scheduling with the objective of minimum operating cost, an intraday rolling optimal scheduling with the objective of minimum intraday economic cost, and a real-time corrective scheduling model with the objective of minimum power fluctuation, respectively. Moreover, it solves the model using alternating direction method of multipliers (ADMMs) to verify the reasonableness and effectiveness of the scheduling strategy proposed in this paper.

2 DC model

Figure 1 shows the basic structure of the IDC system, which is mainly composed of an IDC, a DC microgrid, and a DC operator. The IDC contains the geographical dispersed multiple IDCs as well as the interconnection network, which can provide customers with DC consolidation, business information collaboration, cloud data sharing, data disaster recovery and backup, virtual machine migration, and other conveniences, and realizes flexible expansion of DC scale and flexible business deployment [29]. The DC microgrid contains ES equipment, combustion turbine engines (CTEs) and RES, and other power supply equipment, which serves as a carrier for energy transmission to provide power resources for equipment operation, and meets the power quality requirements of high reliability and stability of various equipment in the DC. The DC operator is a service provider or platform responsible for managing and operating one or more centers. As a representative of all the DCs under its management, it has the right to allocate the data loads in the region under its jurisdiction, and can participate in games and contracts between the DC and other interested parties.

The energy consumption of the DC is modeled according to its energy consumption structure,

$$P_{i,t}^{\text{load}} = P_{i,t}^{\text{server}} + P_{i,t}^{\text{cool}} + P_{i,t}^{\text{other}}, \quad (1)$$

where $P_{i,t}^{\text{load}}$ is the total energy consumption power of the DC, $P_{i,t}^{\text{cool}}$ is the power of DC refrigeration equipment, and $P_{i,t}^{\text{other}}$ is the power for other equipment in the DC.

The server power consumption is modeled based on server utilization

$$P_{i,t}^{\text{server}} = e_{\text{idle}}^{\alpha} m_{i,t}^{\alpha} + \frac{e_{\text{peak}}^{\alpha} - e_{\text{idle}}^{\alpha}}{\mu^{\alpha}} \alpha_{i,t} + e_{\text{idle}}^{\beta} m_{i,t}^{\beta} + \sum_{q \in Q} \frac{e_{\text{peak}}^{\beta} - e_{\text{idle}}^{\beta}}{\mu^{\beta}} \beta_{i,q,t}, \quad (2)$$

where e_{idle}^{α} and e_{peak}^{α} are the no-load power and full-load power of a single server processing interactive workload

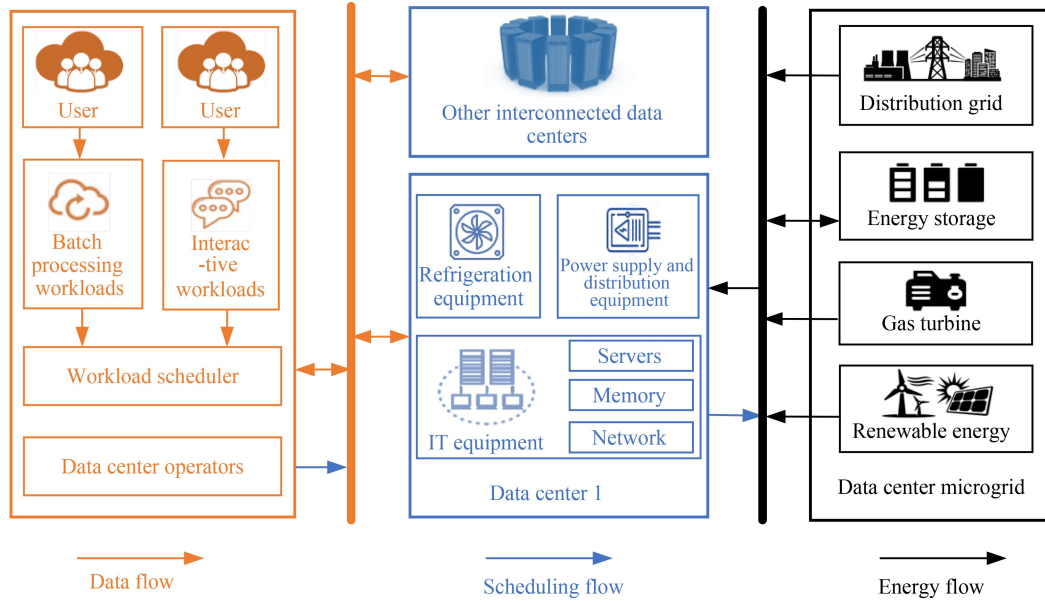


Fig. 1 Basic structure of DC system.

respectively, e_{idle}^{β} and e_{peak}^{β} are respectively no-load power and full-load power of a single server processing batch workload, $\alpha_{i,t}$ is the processing capacity of interactive workloads, $\beta_{i,q,t}$ is the processing capacity of Class q batch workload, $m_{i,t}^{\alpha}$ is the number of servers processing interactive workloads, $m_{i,t}^{\beta}$ is the number of servers processing batch workloads, μ^{α} is the rate at which interactive workloads are handled for a single server, and μ^{β} is the rate at which batch workloads are handled for a single server.

The energy consumption of DC cooling equipment is modeled based on linear principles

$$P_{i,t}^{cool} = \lambda_1 h_{i,t} + e^{cool}, \quad (3)$$

where $h_{i,t}$ is the refrigeration capacity of DC refrigeration equipment, λ_1 is the performance coefficient of the refrigeration equipment, and e^{cool} is the basic energy consumption power of refrigeration equipment.

The indoor temperature change model of the DC is established as

$$T_{i,t}^{in} = T_{i,t-1}^{in} e^{-\frac{\Delta t}{\lambda_2 \lambda_3}} + T_{i,t}^{out} \left(1 - e^{-\frac{\Delta t}{\lambda_2 \lambda_3}}\right) + \lambda_2 \left(1 - e^{-\frac{\Delta t}{\lambda_2 \lambda_3}}\right) (-h_{i,t} + P_{i,t}^{server} + P_{i,t}^{other}), \quad (4)$$

where $T_{i,t}^{in}$ is the indoor temperature of the DC, $T_{i,t}^{out}$ is the outdoor temperature of the DC, and λ_2 and λ_3 are the equivalent thermal parameter of the DC building.

3 Multi-timescale optimal scheduling model for IDCs

3.1 Multi-timescale scheduling framework

The multi-timescale optimal scheduling framework is

divided into day-ahead optimization and intraday optimization, where intraday optimization is divided into rolling optimization and real-time correction, as shown in Fig. 2.

Day-ahead optimized scheduling takes the next day 24 h as the optimization cycle, with a time interval of 1 h, one scheduling plan being developed for each day. Intraday rolling optimization scheduling considers the short-term forecast results of wind and light output, with control time domains t to T as the optimization cycle, with a time interval of 1 h, and one scheduling plan is executed every one hour, and only scheduling plan for time t is executed. Intraday real-time correction scheduling is based on the ultra-short-term prediction results of wind-power with the control time domain $t + n\Delta t$ to $t + 1$ as the optimization period, the time interval of 15 min. It is executed once every 15 min, and only the scheduling plan for the time $t + n\Delta t$ is executed.

3.2 Day-ahead optimal scheduling model

According to the day-ahead predicted values of RES output, workload and environmental parameters, the day-ahead optimal scheduling model of IDC takes the lowest total operation cost as the objective function to develop the next day workload scheduling and energy management strategy, provides reference for day-day operation, and reports the power purchase plan to the power grid company.

3.2.1 Objective function

The objective function of the day-ahead optimal scheduling model of the connected DC is

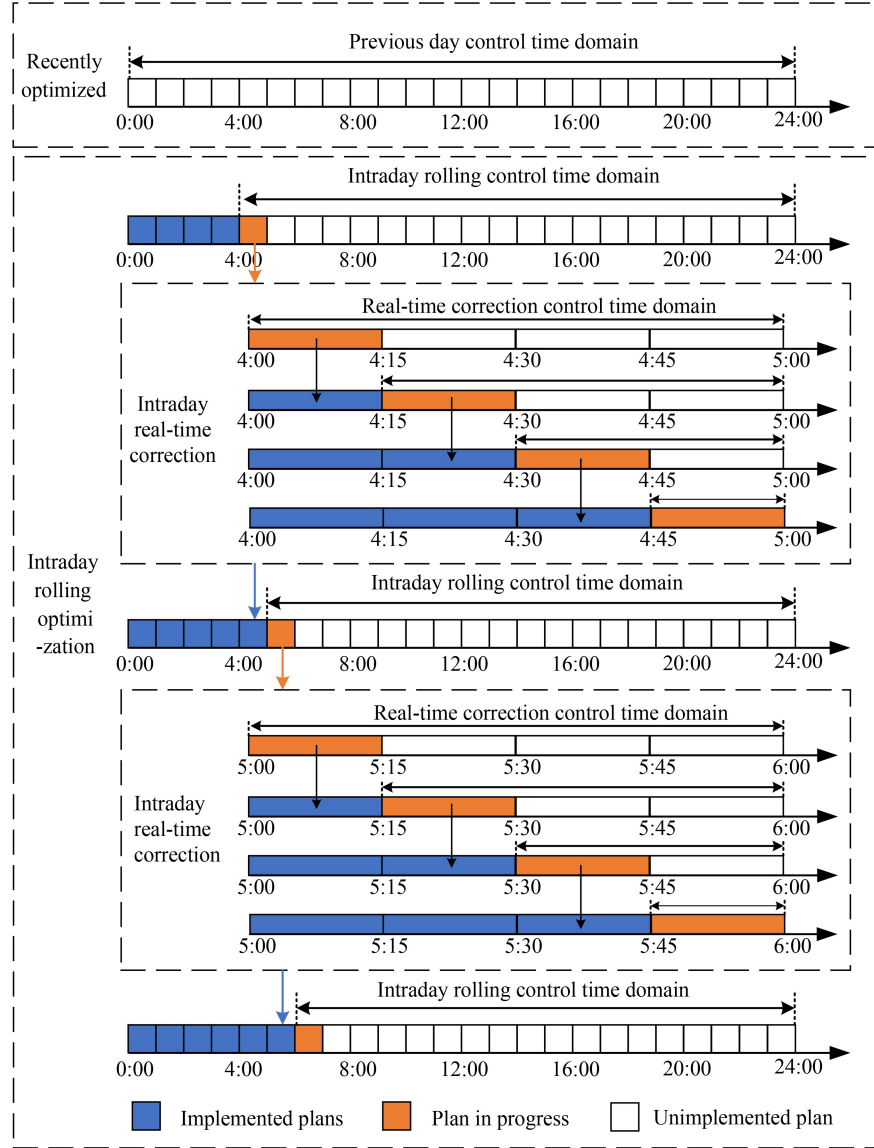


Fig. 2 Multi-timescale optimal scheduling framework.

$$\min \sum_{i \in N} \sum_{t \in T} (C_{i,t}^{\text{grid}} + C_{i,t}^{\text{gas}} + C_{i,t}^{\text{ch}}), \quad (5)$$

where i is the number of the DC, N is the total number of DCs, T is the optimized time period, $C_{i,t}^{\text{grid}}$ is the electricity purchasing cost of the grid at the time t of the DC i , $C_{i,t}^{\text{gas}}$ is the CTE operation cost of the DC, and $C_{i,t}^{\text{ch}}$ is the operation cost of data center energy storage (DCES) equipment.

(1) Power purchase cost of power grid

The calculation formula of power purchase cost of DC network is

$$C_{i,t}^{\text{grid}} = P_{i,t}^{\text{grid}} \gamma_{i,t}^{\text{grid}} \Delta t, \quad \forall t \in T, \forall i \in N, \quad (6)$$

where $P_{i,t}^{\text{grid}}$ is the power consumption of DC grid, $\gamma_{i,t}^{\text{grid}}$ is the power grid electricity price of the region where the DC is located, and Δt is the optimized scheduling time interval.

(2) CTE operation cost

CTEs generate electric energy by burning natural gas, and the relationship between fuel cost and the power of CTEs in DCs can be expressed as

$$C_{i,t}^{\text{gas}} = \frac{\gamma_i^{\text{gas}} P_{i,t}^{\text{gas}} \Delta t}{Q_{\text{LHV}} \eta_{\text{MT}}}, \quad \forall t \in T, \forall i \in N, \quad (7)$$

where γ_i^{gas} is the gas purchase cost per unit power generation of the CTE, $P_{i,t}^{\text{gas}}$ is CTE generating power, Q_{LHV} is the heat value of the CTE, and η_{MT} is the generating efficiency of the CTE.

(3) Operation cost of ES equipment

Energy loss and battery loss will occur in the charging and discharging process of DCES equipment, and the operation cost of ES equipment is

$$C_{i,t}^{\text{ch}} = (P_{i,t}^{\text{bc}} + P_{i,t}^{\text{bd}}) \gamma_i^{\text{ch}} \Delta t, \quad \forall t \in T, \forall i \in N, \quad (8)$$

where $P_{i,t}^{bc}$ and $P_{i,t}^{bd}$ respectively are the charging and discharging power of DCES equipment, and γ_i^{ch} is the unit power usage cost of the ES equipment.

3.2.2 Constraint conditions

(1) Power balance constraints

The power supply in the micro-grid of the DC is optimized to meet the energy demand of the DC, and the RES is optimized to provide electric energy. The power balance constraint conditions are

$$P_{i,t}^{load} = P_{i,t}^{grid} + P_{i,t}^{bd} - P_{i,t}^{bc} + P_{i,t}^{gas} + P_{i,t}^{wind} + P_{i,t}^{solar}, \quad \forall t \in T, \forall i \in N, \quad (9)$$

where $P_{i,t}^{wind}$ is the power generated by the fan in the microgrid of the DC, and $P_{i,t}^{solar}$ is the photovoltaic power generation power in the microgrid of DC.

(2) Constraints of ES equipment

ES equipment should meet the upper and lower limits of charge and discharge power requirements and capacity requirements, whose constraint conditions are

$$0 \leq P_{i,t}^{bd} \leq P^{bd-max}, \quad \forall t \in T, \forall i \in N, \quad (10)$$

$$0 \leq P_{i,t}^{bc} \leq P^{bc-max}, \quad \forall t \in T, \forall i \in N, \quad (11)$$

$$L^{min} \leq L_{i,t} \leq L^{max}, \quad \forall t \in T, \forall i \in N, \quad (12)$$

where P^{bc-max} and P^{bd-max} are respectively the maximum charging and discharging power of the ES device in the DC, η^{bc} and η^{bd} are respectively the charging and discharging efficiency of the ES equipment, $L_{i,t}$ is the capacity of the ES equipment, and L^{min} and L^{max} are respectively the minimum and maximum capacity requirements of the ES equipment.

The capacity of the ES device can be obtained by

$$L_{i,t} = L_{i,t-1} + P_{i,t}^{bc} - P_{i,t}^{bd}, \quad \forall t \in T, \forall i \in N. \quad (13)$$

In addition, the capacity value of the ES device in the initial period and the end period of the scheduling cycle should be consistent, expressed as

$$\begin{aligned} L_{i,t} &= L_{i,0}, \quad \forall t \in T, \forall i \in N, \\ P_{i,t}^{gas-min} &\leq P_{i,t}^{gas} \leq P_{i,t}^{gas-max}, \quad \forall t \in T, \forall i \in N. \end{aligned} \quad (14)$$

(3) Constraint conditions of CTE

The CTE of DC should meet output upper and lower limit constraints and climbing constraints, expressed as

$$P_{i,t}^{gas-min} \leq P_{i,t}^{gas} \leq P_{i,t}^{gas-max}, \quad \forall t \in T, \forall i \in N, \quad (15)$$

$$R_{i,t}^{gas-min} \leq P_{i,t}^{gas} - P_{i,t-1}^{gas} \leq R_{i,t}^{gas-max}, \quad \forall t \in T, \forall i \in N, \quad (16)$$

where $P_{i,t}^{gas-min}$ and $P_{i,t}^{gas-max}$ respectively are the upper and lower limits of CTE output, and $R_{i,t}^{gas-max}$ and $R_{i,t}^{gas-min}$ are respectively the upper and lower limits of CTE climbing power.

(4) Working load constraints

Interactive workload should meet load conservation when scheduling among multiple DCs, whose constraint

conditions are

$$\sum_{i \in N} \alpha_{i,t} = \sum_{i \in N} \varphi_{i,t}^{\alpha}, \quad \forall t \in T, \quad (17)$$

where $\varphi_{i,t}^{\alpha}$ is the interactive work load arriving at DC i at time t .

Batch workloads can be scheduled on a spatiotemporal scale and need to be processed within the deadline. The constraints are

$$\begin{aligned} \sum_{t=1}^{\tau+T_q^{\beta}} \beta_{i,q,t} &\geq \sum_{t=1}^{\tau} \varphi_{i,q,t}^{\beta}, \\ \forall i \in N, \forall q \in Q, \forall \tau \in [1, T - T_q^{\beta}], \end{aligned} \quad (18)$$

$$\begin{aligned} \sum_{t=1}^{\tau} \beta_{i,q,t} &\geq \sum_{t=1}^{\tau} \varphi_{i,q,t}^{\beta}, \\ \forall i \in N, \forall q \in Q, \forall \tau \in [T - T_q^{\beta}, T], \end{aligned} \quad (19)$$

$$\begin{aligned} \sum_{t=1}^{\tau} \beta_{i,q,t} &\leq \sum_{t=1}^{\tau} \varphi_{i,q,t}^{\beta}, \\ \forall i \in N, \forall q \in Q, \forall \tau \in T, \end{aligned} \quad (20)$$

where $\varphi_{i,q,t}^{\beta}$ is the batch work load of Class q arriving at DC i at time t , T_q^{β} is the maximum delay processing time for Class q batch workloads, and τ is any time slot within the scheduling cycle.

(5) Load service level agreement constraints

To ensure the quality of service when DCs process interactive workloads and avoid the long delay caused by queuing, load service level agreement constraints are expressed in Eq. (21) according to the MM1 queuing model

$$\frac{1}{m_{i,t}^{\alpha} \mu^{\alpha} - \alpha_{i,t}} \leq t^{\alpha}, \quad \forall t \in T, \forall i \in N, \quad (21)$$

where t^{α} is the maximum queuing delay allowed when processing interactive workload.

The workload must also meet the server capacity limit when it is processed in the DC, which is expressed as

$$m_{i,t}^{\beta} \geq \frac{\sum_{q \in Q} \beta_{i,q,t}}{\mu^{\beta}}, \quad \forall t \in T, \forall i \in N, \quad (22)$$

$$m_{i,t}^{\alpha} + m_{i,t}^{\beta} \leq m_i^{\max}, \quad \forall t \in T, \forall i \in N, \quad (23)$$

where $m_i^{\max}$ is the maximum number of servers in the DC i .

(6) Constraint conditions of refrigeration system

DC refrigeration power should meet the upper and lower limit constraints

$$P_{i,t}^{cool-min} \leq P_{i,t}^{cool} \leq P_{i,t}^{cool-max}, \quad \forall t \in T, \forall i \in N, \quad (24)$$

where $P_{i,t}^{cool-max}$ and $P_{i,t}^{cool-min}$ respectively are the upper and lower limits of the refrigeration system power.

In the day-ahead scheduling plan, the indoor temperature of the DC refrigeration system should be equal to the set temperature, whose constraint conditions are

$$T_{i,t}^{in} = T^{set}, \quad \forall t \in T, \forall i \in N, \quad (25)$$

where T^{set} is the day-ahead set temperature of the DC room.

(7) Constraints on power transmission of the grid

The transmission power supplied by the power grid to the DC should be less than the maximum allowed transmission power. In addition, this paper does not consider the reverse sale of power from the microgrid in the DC to the power grid. The constraint conditions of the power transmission of the power grid are

$$0 \leq P_{i,t}^{\text{grid}} \leq P^{\text{grid-max}}, \forall t \in T, \forall i \in N, \quad (26)$$

where $P^{\text{grid-max}}$ is the maximum transmission power of the grid supply to the DC.

3.3 Intraday rolling optimized scheduling model

The intraday rolling optimization stage is based on the day-ahead optimization plan and is optimized every hour ahead. Its main function is to adjust the day-ahead scheduling plan to ensure the operation economy of the IDC under the uncertainty of wind-power output, and to provide reference for the intraday real-time correction stage.

3.3.1 Objective function

The rolling optimization model takes the minimum intraday power purchase penalty and operation cost in the time domain of rolling optimization as the objective function. At the same time, in the rolling optimization stage, the constraints on indoor temperature of the DC are relaxed. By using the energy flexibility of the DC refrigeration system, the power consumption is adjusted to reduce the economic loss caused by the prediction error. The objective function of this stage is

$$\min \sum_{i \in N} \sum_{t=t_s}^T (C_{i,t}^{\text{grid-DI}} + C_{i,t}^{\text{gas-DI}} + C_{i,t}^{\text{ch-DI}} + C_{i,t}^{\text{temp-DI}}), \quad (27)$$

where t_s is the rolling optimization start time period, $C_{i,t}^{\text{grid-DI}}$ is the intraday power purchase cost, $C_{i,t}^{\text{gas-DI}}$ is the intraday CTE operating cost, $C_{i,t}^{\text{ch-DI}}$ is the intraday ES equipment operating cost, and $C_{i,t}^{\text{temp-DI}}$ is the indoor temperature offset cost.

The calculation formula of $C_{i,t}^{\text{gas-DI}}$ is the same as Eq. (7), while the calculation formula of $C_{i,t}^{\text{ch-DI}}$ is the same as Eq. (8). The intraday stage power purchase cost is

$$C_{i,t}^{\text{grid-DI}} = (P_{i,t}^{\text{grid-DI}} \gamma_{i,t}^{\text{grid}} + \Delta P_{i,t}^{\text{grid-DI}} \gamma^{\text{grid-DI}}) \Delta t, \quad \forall t \in T, \forall i \in N, \quad (28)$$

$$P_{i,t}^{\text{grid-DI}} = P_{i,t}^{\text{grid}} + \Delta P_{i,t}^{\text{grid-DI}}, \quad \forall t \in T, \forall i \in N, \quad (29)$$

where $\Delta P_{i,t}^{\text{grid-DI}}$ is the intraday purchased power adjustment, and $\gamma^{\text{grid-DI}}$ is the unit penalty cost of the intraday purchased power adjustment.

Indoor set temperature offset cost $C_{i,t}^{\text{temp-DI}}$ is calculated as

$$C_{i,t}^{\text{temp-DI}} = (T_{i,t}^{\text{in-DI}} - T_{i,t}^{\text{in}})^2 \gamma^{\text{temp}} \Delta t, \forall t \in T, \forall i \in N, \quad (30)$$

where $T_{i,t}^{\text{in-DI}}$ is the intraday DC indoor temperature, $T_{i,t}^{\text{in}}$ is the day-ahead indoor temperature, and γ^{temp} is the unit penalty cost of the temperature offset.

3.3.2 Constraint conditions

The DC intraday power balance constraint differs from the day-ahead power balance constraint (according to Eq. (9)), which also includes the intraday power purchase power adjustment amount. The intraday power balance constraint is

$$P_{i,t}^{\text{load-DI}} = P_{i,t}^{\text{grid}} + \Delta P_{i,t}^{\text{grid-DI}} + P_{i,t}^{\text{bd-DI}} - P_{i,t}^{\text{bc-DI}} + P_{i,t}^{\text{gas-DI}} + P_{i,t}^{\text{wind-DI}} + P_{i,t}^{\text{solar-DI}}, \quad \forall t \in T, \forall i \in N. \quad (31)$$

In the process of intraday optimization, DCs relax the restrictions on indoor temperature, indoor temperature meets the upper and lower limits constraints, and the intraday cooling system temperature constraints are

$$T^{\min} \leq T_{i,t}^{\text{in-DI}} \leq T^{\max}, \forall t \in T, \forall i \in N, \quad (32)$$

where $T^{\max}$ is the maximum DC room temperature, and $T^{\min}$ is the minimum DC room temperature.

The ES device constraints, CTE constraints, workload constraints, load service level agreement constraints, and grid transmission power constraints in the day-day rolling optimization model are the same as those in the day-day model, which will not be described here.

3.4 Intraday real-time corrective scheduling model

In the intraday rolling optimization stage, the economy of intraday operation of the IDC is mainly ensured. In the real-time operation process, there is still a small prediction error in the ultra-short-term forecast results of wind-wind output. Therefore, it is necessary to perform real-time correction of large-timescale scheduling plan in intraday rolling optimization on a small-timescale, so that the DC real-time correction phase operation plan tracks the operation plan in the rolling optimization phase, reducing power fluctuations, and thus guaranteeing the economy of the overall scheduling results. The real-time correction is executed every 15 min in advance. According to the results of the RES ultra-short-term prediction, the dispatch plan obtained in the intraday rolling optimization stage is corrected and adjusted. Since the system operation deviation feedback is required in this stage, the optimization model for the real-time correction phase is constructed using an MPC approach.

3.4.1 Objective function

Based on the scheduling plan of rolling optimization stage, each rolling optimization scheduling interval is divided into several intervals with smaller timescales.

According to the change of RES output within the smaller timescale, the minimum power adjustment penalty cost is taken as the objective function which is expressed as

$$\min \sum_{i \in N} \sum_{t=t_s+n\Delta t}^{t_s+1} \|\psi_{i,t}^{\text{RT}} - \psi_{i,t_s}^{\text{DI}}\|_Q^2, \quad (33)$$

where n is the number of time periods within the real-time correction phase, $\psi_{i,t}^{\text{RT}}$ is the output variable of the real-time correction phase, ψ_{i,t_s}^{DI} is the reference value of the output variable, i.e., the t_s time period intraday rolling optimization scheduling plan, and Q is the penalty coefficient matrix of the output variable.

In the real-time correction stage, due to the poor regulation flexibility of the CTE, real-time correction adjustment is not conducted, and it is set that the start-stop and output state of the CTE at this stage should follow the optimal scheduling results of intraday rolling. In addition, due to the coupling characteristics between IDCs in the spatial scheduling of interactive workload, adjusting the workload scheduling plan of a single DC will affect the scheduling plans of other DCs, thus affecting the optimization results of the whole real-time correction stage. Therefore, the interactive workload scheduling plan also follows the results of intraday rolling optimization. Based on this, the output variables of the real-time correction stage mainly include power purchase, the charging and discharging power of ES equipment, and the power consumption of DC load. The formula of output variables is

$$\psi_{i,t}^{\text{RT}} = [P_{i,t}^{\text{grid-RT}}, P_{i,t}^{\text{bd-RT}}, P_{i,t}^{\text{bc-RT}}, P_{i,t}^{\text{load-RT}}]. \quad (34)$$

The control variable is defined as $u_{i,t}^{\text{RT}}$, which is the adjustment amount relative to the rolling optimal dispatch plan, mainly including the purchased power adjustment amount $\Delta P_{i,t}^{\text{grid-RT}}$, the ES equipment charging and discharging power adjustment amount $\Delta P_{i,t}^{\text{bd-RT}}$ and $\Delta P_{i,t}^{\text{bc-RT}}$, and the DC load power adjustment amount $\Delta P_{i,t}^{\text{load-RT}}$, which is given by

$$u_{i,t}^{\text{RT}} = [\Delta P_{i,t}^{\text{grid-RT}}, \Delta P_{i,t}^{\text{bd-RT}}, \Delta P_{i,t}^{\text{bc-RT}}, \Delta P_{i,t}^{\text{load-RT}}], \quad (35)$$

where the first three items $\Delta P_{i,t}^{\text{grid-RT}}$, $\Delta P_{i,t}^{\text{bd-RT}}$, and $\Delta P_{i,t}^{\text{bc-RT}}$ can be directly controlled and adjusted, but the DC power regulation $\Delta P_{i,t}^{\text{load-RT}}$ cannot be directly regulated because it consists of the regulation of the refrigeration system cooling capacity $\Delta h_{i,t}^{\text{RT}}$ and the batch workload adjustment $\Delta \beta_{i,q,t}^{\text{RT}}$. The formula for calculating $\Delta P_{i,t}^{\text{load-RT}}$ is

$$\Delta P_{i,t}^{\text{load-RT}} = \lambda_1 \Delta h_{i,t}^{\text{RT}} + \sum_{q \in Q} \frac{e^{\beta}}{\mu^{\beta}} \Delta \beta_{i,q,t}^{\text{RT}}. \quad (36)$$

The perturbation variable is the predicted variation of the wind-power output in the real-time correction stage within days, which is expressed as:

$$r_{i,t}^{\text{RT}} = [\Delta P_{i,t}^{\text{wind-RT}}, \Delta P_{i,t}^{\text{solar-RT}}]. \quad (37)$$

The update value strategy of the output variable in the real-time correction stage is:

$$\psi_{i,t_s+n\Delta t}^{\text{RT}} = \psi_{i,t_s}^{\text{DI}} + u_{i,t_s+n\Delta t}^{\text{RT}} + r_{i,t_s+n\Delta t}^{\text{RT}}. \quad (38)$$

3.4.2 Constraint conditions

The CTE power in the real-time correction stage is the same as the optimization result in the rolling optimization stage, whose constraint conditions are

$$P_{i,t}^{\text{gas-RT}} = P_{i,t}^{\text{gas-DI}}. \quad (39)$$

The optimization results of the interactive workload scheduling plan in the same rolling optimization stage are

$$\alpha_{i,t}^{\text{RT}} = \alpha_{i,t}^{\text{DI}}. \quad (40)$$

The remaining constraints include intraday power balance constraints, ES device constraints, workload constraints, load service level agreement constraints, and grid transmission power constraints of the same rolling optimization model, which are not listed here.

3.5 Distributed solution method based on ADMM

The centralized optimal scheduling method for IDCs increases the communication and computation burden of the scheduling system, and the optimization results are difficult to describe the interaction process among DCs. Therefore, the alternating direction multiplier method is used to decouple the coupling constraints between IDCs, and the optimal workload interactions are obtained through iteration.

To realize the distributed solution of the original optimization problem, it is necessary to reconstruct the original problem in a distributed manner and divide the original problem into multiple independent subproblems. Since the IDCs have natural partitioning properties, the IDCs are split into multiple individuals to solve separately, that is, the objective function of the optimal total cost of the IDCs is split into the objective function of the optimal cost of multiple independent DCs. The sub-objective function calculation formula f_i for the DC i is defined as

$$\min f_i = \sum_{t \in T} (C_{i,t}^{\text{grid}} + C_{i,t}^{\text{gas}} + C_{i,t}^{\text{ch}}), \text{ s.t. Eqs. (9)–(26)}, \quad (41)$$

The variables of each DC in constraints (Eqs. (6)–(9) and Eqs. (18)–(26)) are independent of each other. Therefore, they can be directly applied to the sub-optimization problem. However, due to the coupling characteristics among multiple DCs in workload space scheduling, decoupling is not possible if the workload space scheduling constraint (Eq. (17)) is relaxed directly. Therefore, the inter-DC interaction variable $\alpha_{i-j,t}^{\text{tr}}$ is introduced, which is defined as the interactive workload volume scheduled between DC i and DC j . When $\alpha_{i-j,t}^{\text{tr}}$ is a positive sign, it indicates that DC i receives the

workload volume scheduled by DC j , and when it is a negative sign, it indicates that DC i schedules the workload volume to DC j . Taking three IDCs as an example, the constraint decoupling process is shown in Fig. 3.

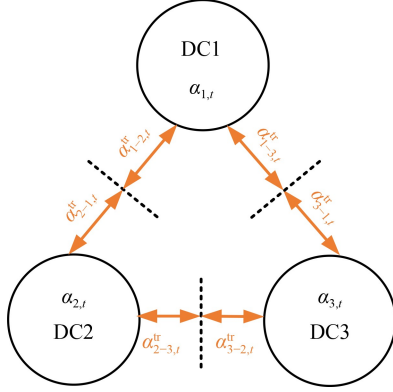


Fig. 3 Schematic diagram of distributed decoupling of IDCs.

After decoupling the constraint, Eqs. (3)–(13) are changed to

$$\alpha_{i,t} = \varphi_{i,t}^\alpha + \sum_{j \in N, j \neq i} \alpha_{i-j,t}^{tr}, \forall t \in T, \forall i \in N. \quad (42)$$

In addition, boundary consistency conditions need to be introduced to ensure convergence of the solution:

$$\alpha_{i-j,t}^{tr} + \alpha_{j-i,t}^{tr} = 0, \forall t \in T, \forall i, j \in N, i \neq j. \quad (43)$$

With the distributed optimization model decoupled, the workload scheduling constraints are relaxed to obtain the augmented Lagrangian function of the DC i , expressed as

$$L_i = \min \left(f_i + \sum_{j \in N, j \neq i} \sum_{t \in T} [\lambda_{i-j,t} (\alpha_{i-j,t}^{tr} + a_{j-i,t}^{tr})] + \sum_{j \in N, j \neq i} \sum_{t \in T} \left(\frac{\rho}{2} \|\alpha_{i-j,t}^{tr} + a_{j-i,t}^{tr}\|_2^2 \right) \right), \quad (44)$$

where $\lambda_{i-j,t}$ is the Lagrange multiplier for the k th iteration between DCs i and j .

Each DC optimizes its own scheduling policy by solving Eqs. (3)–(30), and only interactive workload scheduling information is exchanged between DCs. According to the principle of the alternating direction multiplier method, the form of the $(k+1)$ th iteration update is derived as:

$$\alpha_{i-j,t}^{tr}(k+1) = \operatorname{argmin} L_i(\lambda_{i-j,t}(k), \alpha_{i-j,t}^{tr}(k), \alpha_{j-i,t}^{tr}(k)), \quad \forall i, j \in N, i \neq j, \quad (45)$$

$$\alpha_{j-i,t}^{tr}(k+1) = \operatorname{argmin} L_j(\lambda_{j-i,t}(k), \alpha_{j-i,t}^{tr}(k), \alpha_{i-j,t}^{tr}(k+1)), \quad \forall i, j \in N, i \neq j, \quad (46)$$

$$\lambda_{i-j,t}(k+1) = \lambda_{j-i,t}(k+1) = \lambda_{i-j,t}(k) + \rho(\alpha_{i-j,t}(k+1) + \alpha_{j-i,t}(k+1)), \quad \forall i, j \in N, i \neq j. \quad (47)$$

The convergence criteria are Eqs. (48) and (49), and the iteration is skipped when the conditions are satisfied:

$$\|r_i(k+1)\|_2 = \sum_{j \in N, j \neq i} \sum_{t \in T} \|\alpha_{i-j,t}^{tr} + a_{j-i,t}^{tr}\|_2 \leq \varepsilon_{pri}^1, \quad (48)$$

$$\|s_i(k+1)\|_2 = \sum_{j \in N, j \neq i} \sum_{t \in T} \|\alpha_{i-j,t}^{tr}(k+1) - \alpha_{i-j,t}^{tr}(k)\|_2 \leq \varepsilon_{dual}^1, \quad (49)$$

where ε_{pri}^1 and ε_{dual}^1 are the original residual and pairwise residual convergence metrics, respectively, for the distributed optimal scheduling problem of IDCs.

Since the Lagrange multiplier term in the generalized Lagrange function Eq. (44) is a function related to the workload, the order of magnitude of this term is much larger than that of the original objective function f_i , which will result in the original residuals and pairwise residuals not being able to converge to a smaller value. For this reason, the setting criteria for the original and pairwise residuals should be too small. By referring to Ref. [30], the formula for the convergence index of the original residual and pairwise residual is obtained and expressed as:

$$\varepsilon_{pri}^1 = \sqrt{n} \varepsilon_{abs} + \varepsilon_{rel} \max \left\{ \sum_{j \in N, j \neq i} \sum_{t \in T} \|\alpha_{i-j,t}^{tr}\|_2, \sum_{j \in N, j \neq i} \sum_{t \in T} \|\alpha_{j-i,t}^{tr}\|_2 \right\}, \quad (50)$$

$$\varepsilon_{dual}^1 = \sqrt{n} \varepsilon_{abs} + \varepsilon_{rel} \left(\sum_{j \in N, j \neq i} \sum_{t \in T} \|\lambda_{i-j,t}\|_2 \right), \quad (51)$$

where ε_{abs} and ε_{rel} are the absolute convergence and relative convergence, respectively, whose general value is 0.001.

The two-stage distributed solution process based on the combination of rolling optimization and real-time correction with MPC is shown in Fig. 4.

4 Example analysis

4.1 Multi-timescale scheduling framework

This paper considers an arithmetic analysis of three IDCs managed by one DC operator that are distributed in different regions. The grid tariff uses the grid proxy purchase tariff issued by the local power supply company, see Fig. S1 in Electronic Supplementary Material (ESM), and the scenery output of each DC is shown in Fig. S2. A total of two types of batch workloads in the DC are set, and the ratio of the number of interactive workloads to the two types of batch workloads is set to 6:2:1. Figure S3 shows the arrival rate of each type of workload in the DC. It is assumed that the errors of the intraday rolling optimization phase and the real-time correction phase scenery output follow a normal distribution, and the predicted values of the rolling optimization phase are simulated by superimposing the errors on the predicted values before the day, and the predicted values of the real-time correction phase are simulated by superimposing the errors on the predicted

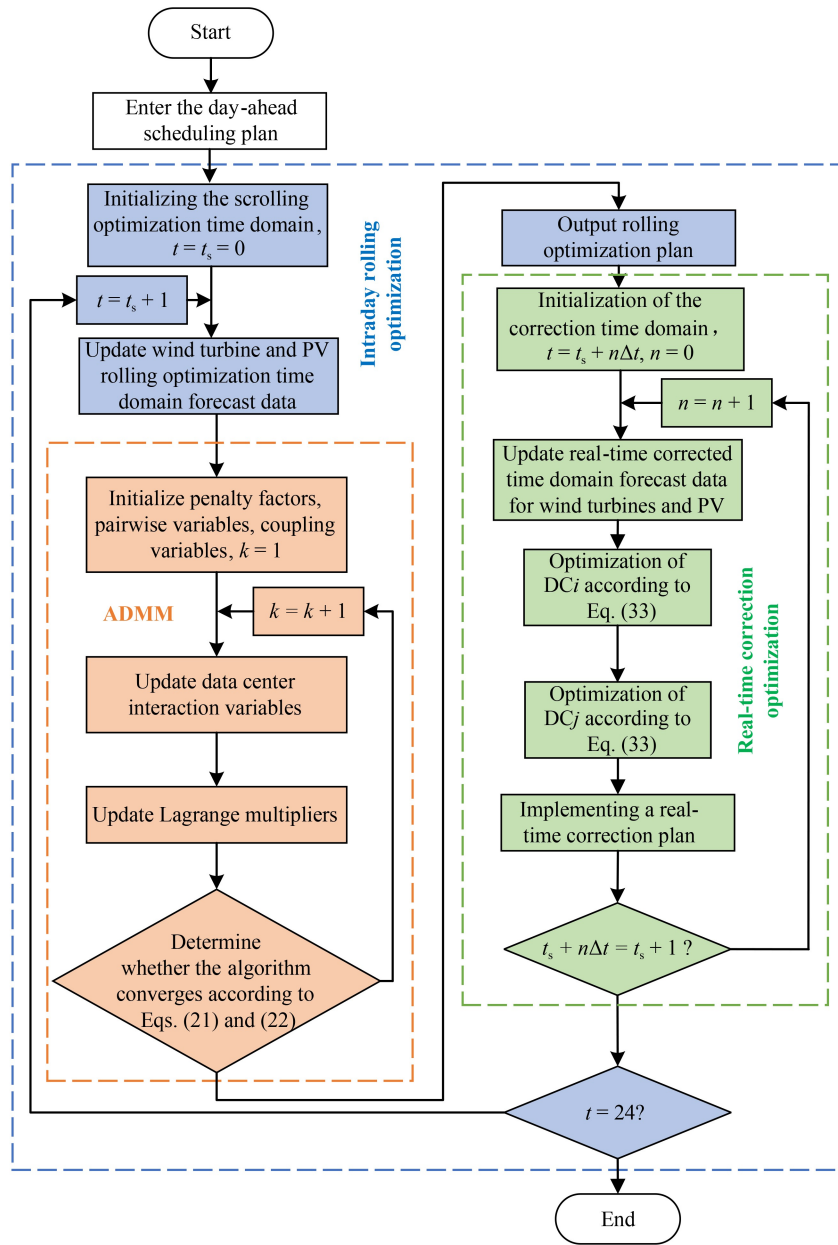


Fig. 4 Multi-timescale distributed optimization solution process.

values of the rolling optimization phase. Figure S4 shows the scenery output of the rolling optimization phase and the real-time correction phase obtained from the simulation. The MATLAB 2019b platform is used to implement the ADMM based multi-timescale optimization programming and call the CPLEX solver to solve the sub-problem. To prove the effectiveness of the method, the following simulation scenarios are set up for comparative analysis:

Scenario 1: Performing rolling optimization phase optimization scheduling with real-time correction phase optimization scheduling using the method proposed in this paper;

Scenario 2: Optimized scheduling in the rolling

optimization phase, without optimal scheduling in the real-time correction phase, with the power fluctuations caused by the scenery power forecast errors compensated by the grid;

Scenario 3: No rolling optimization phase optimization scheduling, no real-time correction phase optimization scheduling, and the power fluctuations caused by the scenery power forecast errors are compensated for by the grid.

4.2 Analysis of optimization results in each scenario

Table 1 shows the operating cost and grid power fluctuation rate of IDCs under different scenarios, and

Fig. 5 shows the grid power fluctuation of IDCs under different scenarios.

From Table 1, it can be seen that the multi-timescale optimal scheduling model (Scenario 1) achieves the lowest grid power fluctuation while ensuring the lowest total operating cost of the IDC, and has the best economy and the lowest power fluctuation rate of all scenarios. Comparing the results of Scenario 1 and the day-ahead optimization, it can be found that Scenario can save US\$6464.5 in economic cost relative to the day-ahead schedule because the intraday optimal scheduling liberalizes the indoor temperature restriction, which allows the DC to change the indoor temperature during the intraday operation and thus reduce the load power cost. The indoor temperature of the cooling system during the intraday real-time correction phase is shown in Fig. S5.

Comparing Scenarios 1 and 2, it can be seen that Scenario 2 bears US\$5430.2 more economic cost than Scenario 1 due to the small-timescale forecast deviation in the real-time correction stage, and the power fluctuation rate of the grid is 5.95% higher than that of Scenario 1. This indicates that the real-time correction scheduling can effectively reduce the impact of small-timescale uncertainty of RES output by making system operation state adjustment based on rolling optimization scheduling. This shows that real-time corrective scheduling can effectively reduce the impact of uncertainty in the small-timescale of RES output, reduce the economic cost, and at the same time achieve effective suppression of power fluctuation of each equipment.

Comparing Scenarios 1 and 3, it can be seen that Scenario 1 has a reduction of US\$3125.2 in the economic cost and a reduction in the power fluctuation

Table 1 Comparison of optimization results of multi-timescale operation under different scenarios

Scenarios	Electricity purchase cost/US\$	CTE/US\$	ES/US\$	Grid power fluctuation rate/US\$
Previous plans	98678.1	33725.2	2236.4	—
Scenario 1	98794.1	26650.9	2730.2	1.29%
Scenario 2	104322.6	26650.9	2631.9	7.24%
Scenario 3	123465.8	33725.2	2236.4	16.52%

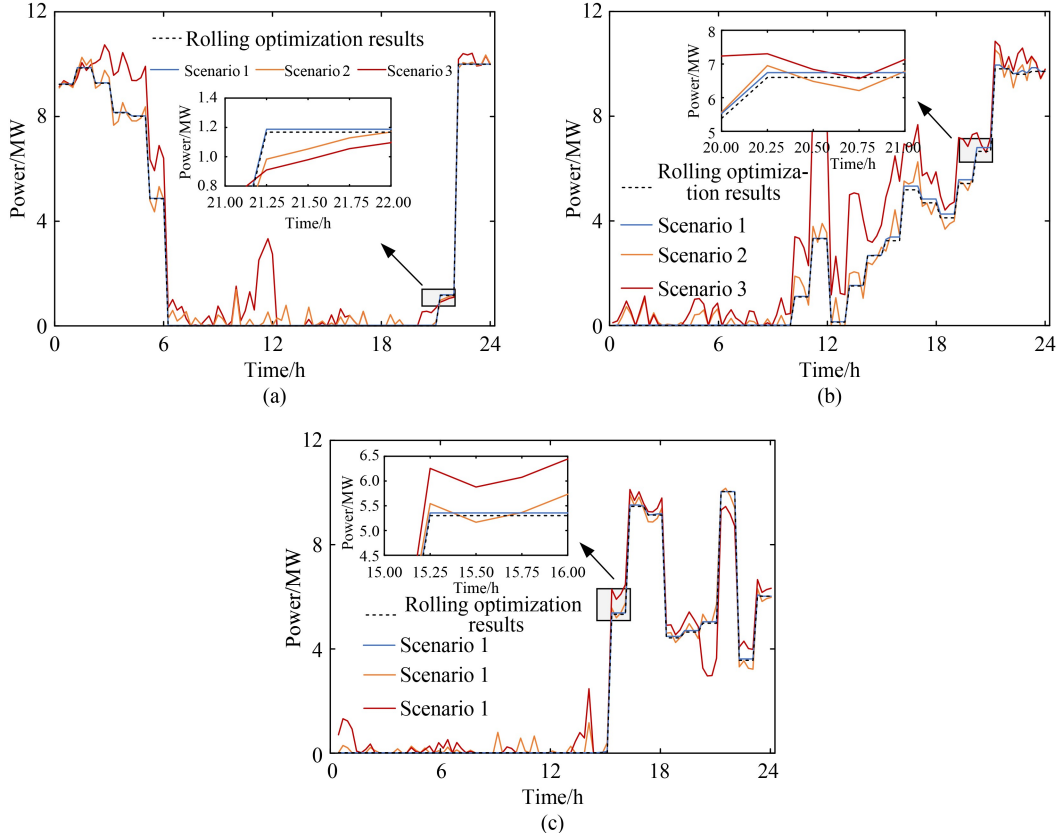


Fig. 5 Power fluctuation of IDC power grid in different scenarios.

(a) DC1 grid power fluctuation; (b) DC2 grid power fluctuation; (c) DC3 grid power fluctuation.

rate of the grid by 15.23% compared with Scenario 3 due to the intraday multi-timescale optimal scheduling, which indicates that the intraday multi-timescale optimal scheduling eliminates the impact of prediction errors on the basis of the day-ahead scheduling plan and greatly ensures the economical operation of the interconnection. The DC operation economy is greatly ensured.

4.3 Rolling optimization scheduling results

Figure 6 shows the energy management optimization results of each DC in the rolling scheduling stage. The priority is to change the power consumption load by adjusting the workload scheduling strategy and regulating the cooling power to eliminate the scenery output forecast deviation in the rolling optimization phase. When the deviation of RES cannot be eliminated by adjusting the power consumption load, the energy management strategy is then adjusted according to the purchase penalty cost and the operating cost of each power supply equipment. Intraday rolling optimization is an economic optimization based on the day-ahead dispatching strategy. To reduce the intraday redispatching cost, the interconnection DC reduces power purchase (e.g., from 1:00 to 2:00 in DC1), reduces CTE generation (e.g., from

14:00 to 17:00 in DC1), increases ES charging (e.g., from 3:00 to 5:00 in DC3) when the intraday forecast of RES output is large and increasing the electricity consumption load (e.g., from 13:00 to 14:00 in DC1), with the opposite scheduling strategy when the intraday forecast of RES output is small as compared to when it is large. In summary, the intraday rolling optimal dispatch achieves economy while filling the large timescale RES forecast deviations with source-load side flexibility resources.

4.4 Real-time correction scheduling results

Figure 7 shows the optimization results of energy management in the IDC during the intraday real-time correction phase. As can be seen from Fig. 7 that, even though the CTEs are not power adjusted and the interactive workloads are not spatially scheduled during the real-time correction phase, the IDCs can still maintain a balanced energy supply for each DC during the real-time correction phase when the RES output fluctuates rapidly by only adjusting the grid power, ES power, and adjusting the batch workload scheduling plan and cooling power.

Figure S6 shows the power fluctuations of each device during the intraday real-time correction phase. As can be

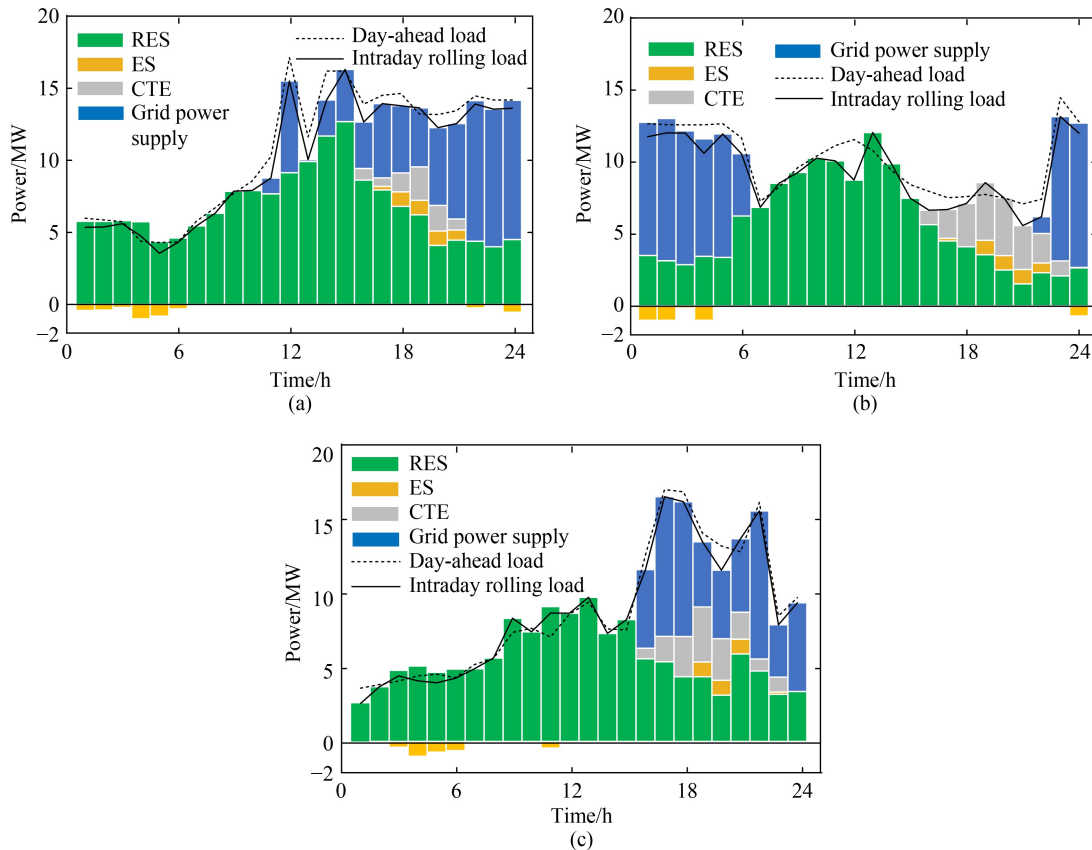


Fig. 6 Energy management optimization results in rolling scheduling phase.

(a) DC1; (b) DC2; (c) DC3.

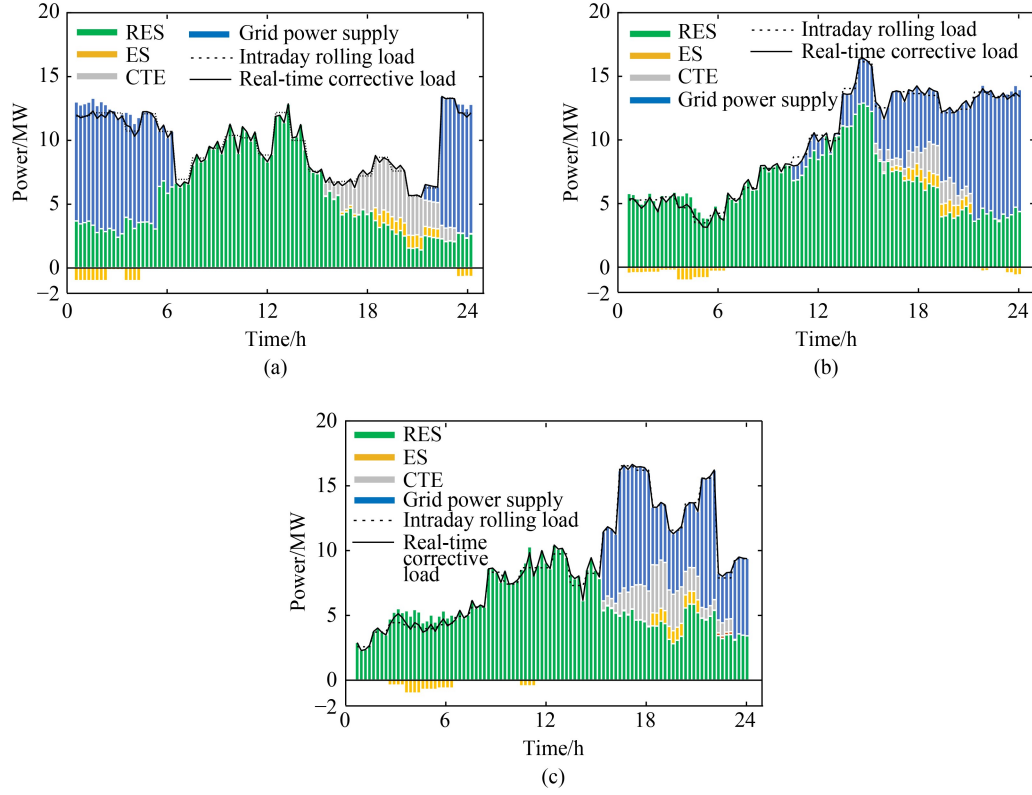


Fig. 7 Energy management optimization results in real-time corrective phase.
(a) DC1; (b) DC2; (c) DC3.

seen from Fig. S6 that the fluctuation of grid power and ES power of each DC is small, while the load power of the DC is relatively more volatile. This is due to the high cost of grid power regulation in intraday operation, and the fact that setting the grid power fluctuation penalty coefficient higher in the real-time correction phase can safeguard the system operating economy. Meanwhile, frequent charge/discharge state changes have a greater impact on the life of the ES battery. Therefore, setting a higher ES power fluctuation penalty factor can extend the life of the ES equipment. Setting a minimum DC load power fluctuation penalty factor can make full use of the DC load-side workload and cooling system scheduling flexibility to perform real-time correction phase power fluctuation leveling. For example, in the 6:00 to 12:00 scenario of DC1, the power prediction error fluctuates greatly, and the corresponding DC load power fluctuation fluctuates greatly in this period, while the grid power and ES power fluctuation are effectively smoothed out.

4.5 Convergence analysis of ADMM

In this paper, the ADMM is used to solve the distributed optimization scheduling problem of interconnected DCs, and the convergence of the objective function values of each DC is shown in Fig. S7. The objective function values of each DC converge to a stable value when the

number of iterations reaches about 20 times. Figure S7(d) compares the optimization results of the centralized algorithm with the optimization results of the ADMM, from which it can be clearly seen that the difference between the ADMM and the centralized algorithm results is smaller. Therefore, the ADMM can converge to the optimal solution within a finite number of iterations when solving the optimal scheduling problem for interconnected DCs.

Calculated according to Eqs. (50) and (51), the convergence criterion for the original residuals and the pairwise residuals is set to 100. The iterative convergence of the residuals of each DC is shown in Fig. 8. When the number of iterations is about 20, the original residuals of each DC have already reached the convergence criterion, and when the number of iterations is about 18, the pairwise residuals of each DC have already reached the convergence criterion. However, as the number of iterations increases, the residuals of each DC still cannot converge to a smaller value below 1. The reason for this is that the coupling variable is the workload, which is of an order of magnitude larger than the operating cost in the objective function. Although the residuals are still unable to converge to a smaller value after reaching the convergence criterion, the impact on the value of the objective function for each DC is minimal, as evidenced by the results of the comparison between the centralized

algorithm and the ADMM in Fig. S7(d).

In this paper, the convergence process of the original residuals of ADMM is compared with that of the Levenberg-Marquardt (LM) in the centralized framework, as shown in Fig. 9. From Fig. 9, it can be seen that compared to the LM algorithm in the centralized framework, which reaches convergence at the 23rd time, the ADMM makes the system computation smaller and faster, and greatly reduces the need for communication and storage. Moreover, each DC only exchanges the workloads after the last iteration, which exchanges less information and protects the information privacy of each DC.

5 Conclusions

In this paper, a multi-timescale optimal scheduling model is proposed for the uncertainty problem of RES output in IDCs. The model consists of day-ahead optimization, rolling optimization, and real-time correction, i.e., the day-ahead optimization phase with the objective of lowest operating cost, rolling optimization phase with the objective of lowest intraday economic cost, and real-time

correction with the objective of lowest power fluctuation. The effects arising from prediction errors are eliminated step by step through coordinated optimization at multi-timescales. The ADMM is used to solve the model. The parameters of the algorithm are set through the relevant reference and the algorithm analysis is analyzed. The main conclusions are as follows:

1) Based on the results of the day-ahead optimal dispatch, the intraday rolling optimization makes use of the source-load side flexibility resources in a rolling manner for intraday economic dispatch, generating an intraday large time-scale dispatch plan, so that the system is able to efficiently consume the RES forecast deviation, and thus safeguard the operating economy of the system.

2) Intraday real-time correction, taking the dispatch results of intraday rolling optimization as a reference, quickly adjusts the workload and the dispatch plan of each equipment, adapts to the small time-scale fluctuation of RES output during online operation, and weakens its uncertain impact. It can suppress power fluctuation while guaranteeing the operation economy, thus ensuring the stable operation of each equipment as well as the power grid.

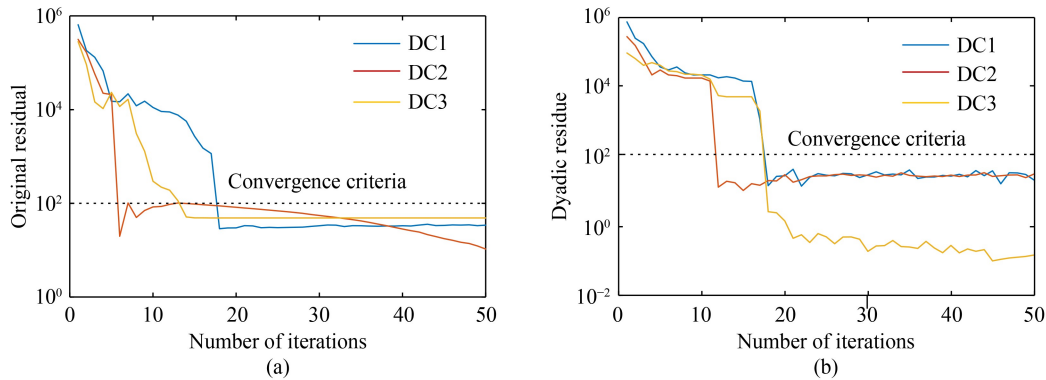


Fig. 8 Convergence of residual iterations across DCs.

(a) Original residuals; (b) dyadic residuals.

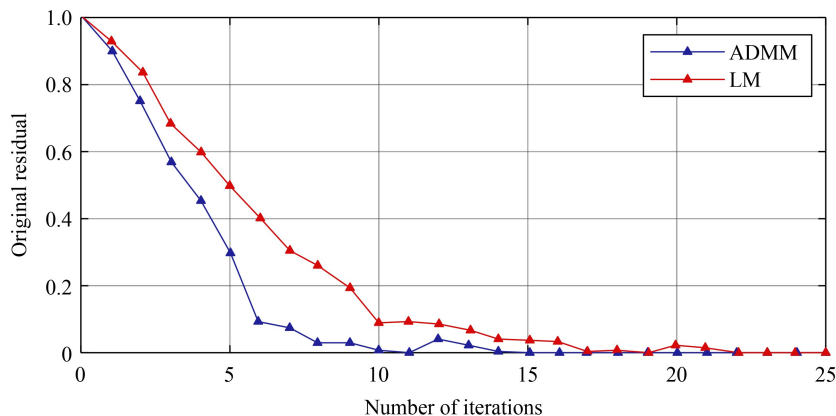


Fig. 9 Original residual convergence process of ADMM and LM.

Competing interests The authors declare that they have no competing interests.

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